

Introduction to Data Science

Data science combines statistics, programming, and domain expertise to extract insights and knowledge from data.

Data Science Process:

1. Data Collection - Gathering raw data
2. Data Cleaning - Handling missing values and outliers
3. Exploratory Analysis - Understanding patterns
4. Feature Engineering - Creating relevant features
5. Modeling - Building predictive models
6. Evaluation - Measuring model performance

Essential tools: Python, Pandas, NumPy, Scikit-learn